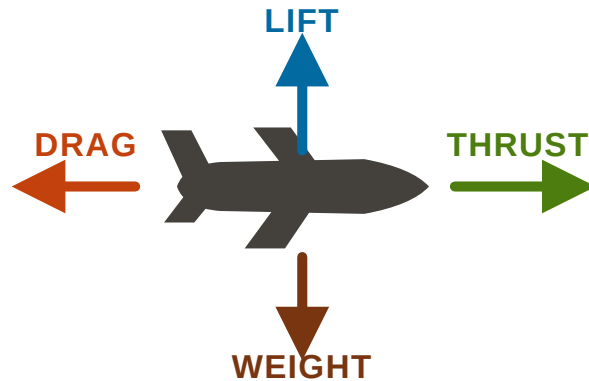


The Four Forces

Every kite. Every boat. Every 777 that rolls out of Everett.



↑ LIFT

Air pushing **UP**. It appears when air moves fast over a curved or tilted surface.

On your kite: the sail tilts into the wind and gets shoved skyward.

↓ WEIGHT

Gravity pulling **DOWN**. Always on. Never turns off. Lift has to beat it.

On your kite: every knot and every inch of tail adds weight. Nothing is free.

→ THRUST

The push **FORWARD**. An engine, a propeller, a paddle — or plain wind.

On your kite: the wind is the engine. On the monster truck, it's the motor.

← DRAG

The push **BACKWARD**. Everything you move through fights you.

On your kite: the tail makes drag *on purpose* — that's what steadies it.

“When he established the force of the wind and measured out the waters...”

Job 28:25

Wind has a force. Water has a measure. We can chart them because they were ordered first — and that is why measuring is worth doing.

Three Big Ideas

Read these out loud on Day 1. Come back to them on Day 10.

1. Air and Water Are Both Fluids

A **fluid** is anything that flows around you. Air is a fluid. Water is a fluid. That is why the kite, the boat and the monster truck belong in one unit — **all three are fighting drag**.

- **Kite:** drag from air
- **Boat:** drag from water — thicker, so it fights harder
- **Truck:** drag from the ground. We call that one **friction**.

A pointy hull and a smooth fuselage solve the exact same problem — that is what Day 8 is really asking.

2. Center of Gravity Is the Balance Point

The one spot where an object's weight is even on every side. Balance a ruler on one finger — your finger is under its center of gravity.

Day 2 test: hold the kite by its bridle. Does it hang level? If the nose dips, the center of gravity sits too far forward — slide the bridle knot back half an inch and hang it again. Write down where it finally balanced.

A 777 has a center of gravity too. Load the cargo wrong and the airplane will not balance, which is why load planners have a whole job.

3. Mama's Jig Rule — How Boeing Runs an Experiment

At Boeing, Mama builds **jigs and fixtures**: tools that hold a part in the **exact same position every single time**, so every part off the line comes out identical. If the jig moved, you could never tell whose fault a bad part was.

A science experiment needs the same discipline. Same ramp, same starting line, same release height, same person on the stopwatch. Then change **one thing only** — and now you know what caused the difference. Scientists call it a **controlled variable**. Mama calls it a jig. Same idea.

How Engineers Write It Down

Three trials, one average, two safety rules.

Three Trials, One Average

One trial is a story. Three trials is data. Run each setup **three times**, write all three numbers, then add them and divide by 3. That number is your **average** — it is what you compare.

Setup	Trial 1	Trial 2	Trial 3	Average
12 in. tail	18.2 s	15.6 s	19.0 s	17.6 s
24 in. tail	31.4 s	29.8 s	33.0 s	31.4 s

$18.2 + 15.6 + 19.0 = 52.8$. Then $52.8 \div 3 = \mathbf{17.6 \text{ seconds}}$. Now you can say the longer tail flew almost **twice** as long — and you can prove it.

New Words — Add These to Your Vocab Page

Weight

Center of Gravity

Fluid

Controlled Variable

Trial

Average

Hypothesis

Bridle

Stabilizer

Inertia is already on your list — that is **Newton's First Law**: an object keeps doing what it is doing until a force changes its mind. Say his name when you use it.

Before You Fly — Two Rules That Are Not Optional

- 1. Never fly a kite near power lines**, and never fly a wet line. If a kite goes into the lines, let go and walk away. It is not worth going back for. Ever.
- 2. Wear a glove on your reeling hand.** Fifty feet of line under load will slice a palm open.